

# CROSS-BASE CORRELATIONS OF ARTIN PRIMES: ENTANGLEMENT, EXCLUSION, AND THE STRUCTURE OF $p - 1$

ABSTRACT. Say a prime  $p$  is *Artin base*  $a$  if  $a$  is a primitive root modulo  $p$ . This paper studies the joint behaviour of Artin status *across bases* at the same prime. We first record a deterministic exclusion law: for *any* two bases  $a, b$ , writing  $d = \text{sqf}(ab)$  for the squarefree part of  $ab$ , no prime  $p$  with  $\left(\frac{d}{p}\right) = -1$  is simultaneously Artin for  $a$  and for  $b$ . That condition is a union of residue classes modulo  $4d$ , so whenever  $d > 1$  — equivalently, whenever  $a$  and  $b$  do not have the same squarefree part — half of all primes are barred outright; the proof is two lines from the multiplicativity of quadratic characters. A corollary is the triple form of the law: if  $\text{sqf}(c) = \text{sqf}(ab)$  then no odd prime is simultaneously Artin for  $a$ ,  $b$  and  $c$  — for instance, no odd prime has 2, 5 and 10 all as primitive roots. Over all 66 pairs of the base set below and all 50,847,524 primes below  $10^9$ , the number of primes violating the pair law is zero, as it must be. We then measure, for all 66 pairs from the twelve bases  $\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}$  and all 50,847,532 primes up to  $10^9$ , the same-prime correlation  $\phi(a, b)$  between the indicator variables of Artin status. Three empirical findings emerge. First, the correlation is positive for 64 of the 66 pairs (mean  $\phi = 0.0352$ , individual  $|z|$  up to 476 — which reflects the  $5 \times 10^7$  sample size rather than a large effect; the correlations explain about 0.1% of variance), despite the exclusion law: the common cause  $p - 1$  dominates. The only negative pairs are (2, 10) and (3, 15), the two for which the barring character has conductor 5, the smallest occurring. Second, conditioning on the number of prime factors  $\omega(p - 1)$  removes only 55% of the mean correlation, but conditioning on a finer signature of  $p - 1$  — the 2-adic valuation, the divisibility of  $p - 1$  by each prime up to 13, and the number of larger factors — removes 95%: the cross-base correlation is, almost entirely, a reflection of the fact that all bases see the same factorisation of  $p - 1$ . Third, the residual that survives fine conditioning is concentrated exactly on the multiplicatively dependent pairs (mean absolute residual 0.018 within dependent pairs, or 0.008 excluding one large positive outlier, versus 0.001 for multiplicatively independent pairs), and is localised in the quadratic-character channel — identifying it as the statistical residue of the exclusion law itself, visible once the common cause is conditioned away. Finally we evaluate the joint densities predicted by the Matthews–Moree–Stevenhagen Kummer-degree heuristic for all 66 pairs: with no fitted parameters they match the measured densities to a mean relative error of 0.012%, and they reproduce the sign reversal on the entangled pairs. The deterministic exclusion law and the statistical residual are thereby identified as two faces of one object: the quadratic character  $\chi_{\text{sqf}(ab)}$ , appearing pointwise as a congruence obstruction and asymptotically as a degree collapse.

## 1. INTRODUCTION

Artin’s conjecture on primitive roots concerns one base at a time: for non-square  $a \neq -1$ , the set

$$A_a = \{p \text{ prime} : a \text{ is a primitive root mod } p\}$$

should have density given by a rational multiple of Artin’s constant, as proved under GRH by Hooley [4]. The joint distribution for *several* bases at the same prime was treated by Matthews [11], who computed (under GRH) the density of primes for which several prescribed integers are simultaneously primitive roots. Lenstra’s Galois-theoretic framework [8] covers a

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single finitely generated group  $W$  with a prescribed index condition, and Lenstra explicitly lists the simultaneous problem — “the set of primes  $q$  for which both 2 and 3 are primitive roots” (his (8.17), citing Matthews) — among the sets to which his results do not directly apply. The systematic Kummer-degree treatment of the simultaneous densities is due to Moree–Stevenhagen [10], whose character-sum method reproves Matthews’ theorem; see also Moree’s survey [12]. The qualitative message of that theory is that the events  $p \in A_a$  and  $p \in A_b$  are *not* independent, and that the deviation from independence is governed by the interaction of the Kummer–Galois extensions attached to  $a$  and  $b$  — in modern language, by *entanglement* between them.

That framework remains actively developed. Järvinen–Perucca–Sgobba [5] give a unified treatment of Artin-type problems for finitely generated groups in number fields; Kimmel [6] treats number-field and matrix variants (all results there conditional on GRH); Klurman–Shparlinski–Teräväinen [7] obtain results on average over the base; Sgobba [13] establishes unconditional results in cases where GRH can be circumvented; and Goldmakher–Martin–Péringuey [3] propose refined conjectures on the distribution of the multiplicative order, which bear on the structure of  $p - 1$  that plays the central role in Section 5 below.

The density theory for several bases at a fixed prime is well developed: [5] treats, under GRH, the joint index map  $\mathbf{p} \mapsto (\text{Ind}_{\mathbf{p}}(W_1), \dots, \text{Ind}_{\mathbf{p}}(W_n))$  in full generality, of which the simultaneous primitive root problem over  $\mathbb{Q}$  is the special case  $H = \{(1, \dots, 1)\}$ . What appears not to have been done is the empirical side: measuring these joint densities and the resulting same-prime correlations at scale, and decomposing them by conditioning on the factorisation of  $p - 1$ . That is our subject; we take [11, 10] as the source of the density predictions tested in Section 7. As those predictions are conditional on GRH, so is every comparison we make with them; our measurements themselves are of course unconditional.

This paper is the third in a series studying correlations of Artin status computationally at the scale of  $10^9$  [1, 2]. The first paper [1] measured the correlation between the Artin statuses (base 10) of *consecutive* primes and decomposed it into residue channels; the second gives a classification (companion preprint, not yet refereed) of deterministic exclusion laws for consecutive Artin primes in arbitrary bases and identified the conductor of the quadratic character  $\chi_a$  as the parameter governing the statistical correlation. Here we complete the picture along the remaining axis: *fix the prime, vary the base*.

Two questions organise the paper.

(Q1) *What is deterministically forbidden?* For consecutive primes, the answer given in [2] was a set of gap classes on which the quadratic character is forced to flip. For a single prime and several bases, the answer is a congruence condition on  $p$  that bars joint Artin status on half of all primes, for every pair of bases not sharing a squarefree part:

**Theorem 1** (Same-prime pair exclusion; see Theorem 3). *Let  $a, b$  be positive integers and set  $d = \text{sqf}(ab)$ . Then no odd prime  $p$  with  $\left(\frac{d}{p}\right) = -1$  is simultaneously Artin base  $a$  and Artin base  $b$ .*

By quadratic reciprocity the excluded set is an explicit union of residue classes modulo  $4d$ , and whenever  $d > 1$  it has density exactly  $1/2$  among the odd primes. So for any two bases with  $\text{sqf}(a) \neq \text{sqf}(b)$  — with no requirement that they sit in a special configuration, and with no third base in sight — half the primes are deterministically forbidden from being Artin for both. (If  $\text{sqf}(a) = \text{sqf}(b)$  then  $d = 1$ , the character is principal and the statement is vacuous; see Section 2.) For example 2 and 6 are never simultaneously primitive roots for  $p \equiv 5, 7 \pmod{12}$ ; 2 and 10 never for  $p \equiv 2, 3 \pmod{5}$ .

Specialising the barring character to a third base of the set gives the triple form, which was our original route to the law:

**Corollary 2** (Triple exclusion; see Theorem 4). *Let  $a, b, c$  be positive integers with  $\text{sqf}(c) = \text{sqf}(ab)$ . Then no odd prime is simultaneously Artin base  $a$ , Artin base  $b$ , and Artin base  $c$ .*

For example, no odd prime has all of 2, 5, 10 as primitive roots; none has all of 3, 5, 15; and, since  $\text{sqf}$  is all that matters, none has all of 2, 5, 40. (The restriction to odd  $p$  is essential and not cosmetic: see the sharpness remark below.) Like the gap exclusion law of [2], both statements are elementary: they need only the multiplicativity of the quadratic character and the fact that a quadratic residue is not a primitive root. The triple criterion is moreover classical in the density theory, where the set of primes for which  $a_1, \dots, a_n$  are simultaneously primitive roots is known to be empty precisely when some odd-size subset  $I$  has  $\prod_{i \in I} a_i$  a square (see [10, §5, (5.2)]; the criterion is originally Matthews' [11]), of which Theorem 4 is the case  $n = 3$ ,  $I = \{a, b, c\}$ . What we add is the elementary pointwise proof, the sharpening to odd  $p$  with no further hypotheses, and — the point of emphasis here — the observation that the third base is inessential: the obstruction is a congruence, of density 1/2, attached to the single character  $\chi_{\text{sqf}(ab)}$ . That character is the object the rest of the paper measures.

(Q2) *What does the joint distribution actually look like, and what explains it?* Here the data speak. We compute the exact  $2 \times 2$  contingency table of  $(\mathbf{1}[p \in A_a], \mathbf{1}[p \in A_b])$  over all 50,847,532 primes  $p \leq 10^9$ , for all 66 pairs from the base set

$$\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\},$$

chosen to contain many multiplicative triples (15 of the 66 pairs lie in a triple  $\text{sqf}(ab) = \text{sqf}(c)$  within the set) alongside multiplicatively independent pairs. The findings, developed in Sections 4–6:

- (1) *The exclusion law is visible exactly.* For every one of the 66 pairs, the number of primes  $p \leq 10^9$  that are Artin for both  $a$  and  $b$  while  $\chi_a(p) \neq \chi_b(p)$  is 0, as Theorem 3 requires; on the complementary half of the primes the joint density is 0.299. The deterministic law is thus not merely proved but exhibited, on 50,847,524 primes and  $3.4 \times 10^9$  pair-prime incidences.
- (2) *Positivity, nonetheless.*  $\phi(a, b) > 0$  for 64 of 66 pairs, with mean 0.0352 and  $z$ -scores up to 476: knowing that  $a$  is a primitive root mod  $p$  makes it *more* likely that  $b$  is, even though half of all primes are barred from having both. The common cause  $p - 1$  outweighs the exclusion. The largest correlations all involve base 5 (e.g.  $\phi(5, 13) = 0.0667$ :  $P(A_{13} \mid A_5) = 0.416$  against  $P(A_{13} \mid \neg A_5) = 0.350$ ).
- (3) *Where the exclusion wins.* The only two negative correlations are  $\phi(2, 10) = -0.0302$  and  $\phi(3, 15) = -0.0303$ . Both are multiplicatively dependent, but dependence alone is not the discriminant: fifteen of the 66 pairs are dependent and thirteen of those are positively correlated. What singles out these two is the conductor of the barring character  $\chi_{\text{sqf}(ab)}$ : both have  $\text{sqf}(ab) = 5$ , of conductor 5 — the smallest conductor occurring among the dependent pairs, and the only odd prime one. A small conductor is what lets the exclusion of Theorem 3 survive the averaging over  $p - 1$  instead of being diluted by it.
- (4) *The positive correlation is  $p - 1$  structure, not entanglement.* Conditioning on  $\omega(p - 1)$  removes 55% of the mean correlation; conditioning on a fine signature of  $p - 1$  — namely  $(\min(v_2(p - 1), 4), \text{the set of primes } q \leq 13 \text{ dividing } p - 1, \min(\omega_{>13}(p - 1), 3))$  — removes 95%, leaving a mean residual of 0.0019. All bases are tested against the same integer  $p - 1$ ; a prime whose  $p - 1$  is “primitive-root friendly” (few small factors, small

- 2-part) is friendlier for *every* base simultaneously. This common-cause structure, not any interaction between the bases, is the bulk of the correlation.
- (5) *What survives is entanglement, and it is exactly localised.* After fine conditioning, the residual correlation among multiplicatively *dependent* pairs (mean absolute residual 0.0179, or 0.0077 excluding the single large positive outlier (2, 6), with individual values reaching  $-0.034$  for (2, 10) and  $-0.026$  for (5, 15)) is eighteen times that of independent pairs (0.0010), or eightfold with (2, 6) excluded; and additionally conditioning on the quadratic-residue statuses  $(\chi_a(p), \chi_b(p))$  flattens dependent and independent pairs to a common  $+0.005$ . The residual is thus identified, empirically, as the quadratic-character entanglement  $\chi_a \chi_b = \chi_{\text{sqf}(ab)}$  — the statistical shadow of Theorem 3.
  - (6) *Theory reproduces all of it.* Evaluating the Matthews–Moree–Stevenhagen Kummer-degree heuristic (3) for each pair (Section 7) predicts the measured joint densities to a mean relative error of 0.012% over the 66 pairs, with no fitted parameters, including the negative  $\phi$  of the two entangled pairs. The entangled degree collapse in the model is precisely the character relation behind Theorem 3.

The theme of the series persists: at every level, the joint behaviour of Artin primes decomposes into a small deterministic core (here, the pair exclusion law) plus a statistical component with an identifiable arithmetic carrier (here, the shared factorisation of  $p - 1$ , with entanglement as a measurable second-order correction). What this paper adds to the pattern is that the deterministic core and the statistical carrier are the *same* character seen twice:  $\chi_{\text{sqf}(ab)}$  bars half the primes pointwise, and reappears as the degree collapse that bends the predicted densities.

## 2. THE SAME-PRIME EXCLUSION LAWS

Throughout, for a positive integer  $a$  we write  $d_a = \text{sqf}(a)$  for its squarefree part and  $\chi_a = \left(\frac{d_a}{\cdot}\right)$  for the quadratic character attached to  $\mathbb{Q}(\sqrt{a})$ ; thus  $\chi_a(p) = +1$  exactly when  $a$  is a quadratic residue modulo  $p$  (for  $p \nmid 2a$ ). We allow  $a$  to be a square, in which case  $d_a = 1$  and  $\chi_a \equiv +1$  is the principal character; this convention costs nothing and lets the statements below dispense with a non-squaredness hypothesis. We have

$$p \in A_a \implies \chi_a(p) = -1, \quad (1)$$

since a quadratic residue generates a subgroup of index at least 2 in  $(\mathbb{Z}/p\mathbb{Z})^\times$ .

Everything in this section follows from one line: the quadratic character is completely multiplicative, so for any positive integers  $a, b$  and any odd prime  $p \nmid ab$ ,

$$\chi_a(p) \chi_b(p) = \left(\frac{d_a d_b}{p}\right) = \left(\frac{\text{sqf}(ab)}{p}\right) = \chi_{\text{sqf}(ab)}(p), \quad (2)$$

since  $d_a d_b$  and  $\text{sqf}(ab)$  differ by a square. Combining (2) with the obstruction (1) gives the following.

**Theorem 3** (Pair exclusion). *Let  $a, b$  be positive integers and set  $d = \text{sqf}(ab)$ . Then for every odd prime  $p$  with  $\left(\frac{d}{p}\right) = -1$ , the prime  $p$  is not simultaneously Artin base  $a$  and Artin base  $b$ .*

*Proof.* If  $p \mid ab$  then one of the two bases is  $\equiv 0 \pmod{p}$  and so is not a primitive root; assume therefore  $p \nmid ab$ . By (2),  $\chi_a(p) \chi_b(p) = \left(\frac{d}{p}\right) = -1$ , so  $\chi_a(p)$  and  $\chi_b(p)$  have opposite signs and in particular one of them equals  $+1$ ; by (1) the corresponding base is not a primitive root.  $\square$

The condition  $\left(\frac{d}{p}\right) = -1$  is a condition on residue classes. For  $d > 1$  the character  $\left(\frac{d}{\cdot}\right)$  is non-principal, of fundamental discriminant  $d$  if  $d \equiv 1 \pmod{4}$  and  $4d$  otherwise; either way the discriminant divides  $4d$ , so the excluded set is an explicit union of reduced classes modulo  $4d$ , and exactly half of those classes carry character  $-1$ . By the prime number theorem in

arithmetic progressions the excluded set then has density exactly  $1/2$  among the odd primes. So for any two bases with  $d > 1$  — no third base required, no special configuration — half of all primes are deterministically barred from being Artin for both, and Theorem 3 is a *congruence* exclusion law.

The excluded case  $d = 1$  is exactly  $\text{sqf}(a) = \text{sqf}(b)$ , i.e.  $ab$  a perfect square, i.e.  $\chi_a = \chi_b$ : then  $\chi_a \chi_b = \chi_a^2 \equiv +1$ , nothing is ever barred, and Theorem 3 is vacuous rather than false. (For instance  $a = 2$ ,  $b = 8$ : the two bases generate the same quadratic field, and indeed 2 and 8 are simultaneously primitive roots for  $p = 3, 5, 11, 29, 53, \dots$ ) Such pairs are precisely the ones our base set avoids: all twelve bases are squarefree and distinct, so  $d > 1$  for all 66 pairs. Writing the classes out in the four cases we return to below:

$$\begin{aligned} (2, 6), d = 3: p \equiv 5, 7 \pmod{12}; & \quad (3, 6) \text{ and } (5, 10), d = 2: p \equiv 3, 5 \pmod{8}; \\ (2, 10), d = 5: p \equiv 2, 3 \pmod{5}. \end{aligned}$$

We verified each over all primes below  $6 \times 10^4$  with no exceptions, and Section 4 verifies the theorem itself against all 50,847,524 primes below  $10^9$  for all 66 pairs of our base set simultaneously. Theorem 3 is also machine-checked in Lean 4 against Mathlib, as is the  $(5, 10)$  instance above with its barring condition discharged from  $p \bmod 8$ ; see `LEAN_NOTE.md` in the accompanying package.

The triple form of the law, which was our original route to it, is the special case in which the barring character is supplied by a third base of the set.

**Theorem 4** (Triple exclusion). *Let  $a, b, c$  be positive integers with  $\text{sqf}(c) = \text{sqf}(ab)$  and let  $p$  be an odd prime. Then  $p$  is not simultaneously Artin base  $a$ , Artin base  $b$ , and Artin base  $c$ .*

*Proof.* If  $p$  is Artin base  $c$  then  $\chi_c(p) = -1$  by (1), and  $\chi_c = \chi_d$  with  $d = \text{sqf}(ab) = \text{sqf}(c)$ ; now apply Theorem 3.  $\square$

**Corollary 5.** *Within any multiplicative triple — for instance  $(2, 5, 10)$ ,  $(3, 5, 15)$ ,  $(3, 7, 21)$ ,  $(2, 3, 6)$  — a prime can be Artin for at most two of the three bases.*

The gain in passing from Theorem 4 to Theorem 3 is not cosmetic, though it is a gain in usable content rather than in logical strength: both are pointwise statements at every odd prime. The difference is in what one must know to apply them. To invoke Theorem 4 one must first identify a third base  $c$  with  $\text{sqf}(c) = \text{sqf}(ab)$  and establish that  $c$  is a primitive root at the prime in question — itself an instance of the Artin problem, conjectural in general and (under GRH) of positive but base-dependent density. Theorem 3 needs neither: the barring condition is a congruence, decidable from  $p$  alone by reciprocity, and for  $d > 1$  it holds on a set of density exactly  $1/2$ . The triple law is the shadow that the pair law casts when one insists on reading the barring character off a primitive root instead of off a residue class.

**Remark 6** (Sharpness of the hypotheses). In both theorems only “ $p$  odd” is needed. Non-squareness of the bases is not required: if  $a$  is a square then  $\chi_a(p) = +1$  and  $a$  is not a primitive root, so the conclusion is immediate. The same remark disposes of  $c = 1$  in Theorem 4, which arises whenever  $\text{sqf}(a) = \text{sqf}(b)$ . The hypothesis  $p \neq 2$  must however be kept, and is not removable:  $(\mathbb{Z}/2\mathbb{Z})^\times$  is trivial, so every odd integer is vacuously a primitive root mod 2 and e.g.  $(a, b, c) = (3, 5, 15)$  violates the conclusion there. The accompanying Lean formalisation (Section 3) confirms independently that the  $a \not\equiv 0$  and  $b \not\equiv 0 \pmod{p}$  assumptions are never used.

**Remark 7.** The exclusion is sharp: primes Artin for exactly two of  $(2, 5, 10)$  exist in abundance. The smallest is  $p = 3$ , where 2 and 5 are both primitive roots (each of order  $2 = p - 1$ ) while  $10 \equiv 1$  is not; the smallest for which 10 itself is Artin is  $p = 7$ , where 5 and 10 are both primitive

roots (each of order 6) while 2 has order 3; likewise  $p = 23$ , where 5 and 10 are primitive roots and 2 is a quadratic residue of order 11. More systematically, in our data the pair (2, 10) has  $P(A_{10} \mid A_2) = 0.355$ : two-of-three is common, three-of-three is impossible.

**Remark 8.** Theorem 3 is the same-prime analogue of the gap exclusion law for consecutive primes proved in [2]: both are instances of a parity obstruction in the group generated by the quadratic characters involved, and in both the excluded set is a union of residue classes. There the two characters were  $\chi_a$  evaluated at  $p$  and at  $p + g$ , and the obstruction was arranged by the shift  $g$ ; here the characters belong to different bases at the same prime, and the obstruction is arranged by multiplicative dependence. The Galois-theoretic density computations of Matthews [11] and Moree–Stevenhagen [10] contain this relation as the vanishing criterion recalled in Section 1 (the degree of the compositum of the relevant Kummer extensions drops); the elementary pointwise form seems worth recording.

### 3. DATA AND COMPUTATION

By a segmented sieve we enumerated all primes  $p \leq 10^9$  (50,847,534 primes). The programs discard only the smallest primes, for which the 128-bit Euler-criterion arithmetic and the signature encoding are degenerate:  $p < 5$  for the correlation and decomposition runs, leaving 50,847,532 primes, and  $p < 31$  for the signature-and-character run of Section 6, leaving 50,847,524. A prime dividing a base is *not* discarded; it is simply recorded as non-Artin for that base, which is correct, since 0 is not a primitive root. The two counts therefore differ by exactly the eight primes in  $[5, 31)$ . For each prime we factored  $p - 1$  once by trial division and computed the Artin indicator for each of the twelve bases by the standard criterion ( $a^{(p-1)/\ell} \not\equiv 1 \pmod p$  for every prime  $\ell \mid p - 1$ , in 128-bit arithmetic). One streaming pass accumulates, for each of the 66 unordered base pairs, the  $2 \times 2$  contingency table of joint Artin status — globally, and stratified by (i)  $\omega(p - 1)$  and (ii) the fine signature

$$\text{sig}(p) = \left( \min(v_2(p - 1), 4), \{q \leq 13 : q \mid p - 1\}, \min(\omega_{>13}(p - 1), 3) \right),$$

a partition of the primes into at most  $4 \cdot 32 \cdot 4 = 512$  cells (for odd  $p$  we always have  $v_2(p - 1) \geq 1$ , so the truncated valuation takes four values, not five). The signature records exactly the information about  $p - 1$  that the Artin criterion consults for small bases: the 2-part and the small odd prime divisors. A separate run, also to  $10^9$  (50,847,524 primes), stratified each pair additionally by the quadratic-residue statuses  $(\chi_a(p), \chi_b(p))$  of the two bases on top of the signature. The whole computation runs in under two hours on a single core of a commodity machine (2-core Intel Xeon E5-2673 v4, 8 GB RAM); all counts are exact.

As validation, the marginal Artin densities agreed with the Hooley densities to within  $6 \times 10^{-5}$  absolute (0.014% relative) for every base — e.g. 0.373957 for base 2, against Artin’s constant 0.373956, and 0.393642 for base 5, whose density carries the classical correction factor  $C \cdot \frac{20}{19}$  for  $d \equiv 1 \pmod 4$  (see [4, 12]).

### 4. THE CROSS-BASE CORRELATION

For each pair write  $\phi(a, b)$  for the Pearson correlation of the indicator variables (the  $\phi$  coefficient of the  $2 \times 2$  table). At  $N \approx 5.08 \times 10^7$  primes, the standard error of  $\phi$  under independence is  $N^{-1/2} \approx 0.00014$ ; as in the previous papers of the series, the enormous resulting  $z$ -scores certify only that sampling noise is negligible, and the relevant quantity is the effect size. The 66 pairs are not independent tests — twelve bases generate them and all share the same primes — but the smallest  $|z|$  among them is 169, against a Bonferroni threshold of 3.4 at the 5% level, so no multiple-comparison correction affects any conclusion below.

TABLE 1. Same-prime cross-base Artin correlation at  $x = 10^9$ : summary and extremes. “Dependent” means the pair lies in a multiplicative triple within the base set ( $\text{sqf}(ab) = \text{sqf}(c)$ ).

|                              | pairs | mean $\phi$ |
|------------------------------|-------|-------------|
| all pairs                    | 66    | +0.0352     |
| multiplicatively independent | 51    | +0.0363     |
| multiplicatively dependent   | 15    | +0.0314     |
| largest: (5, 13)             |       | +0.0667     |
| smallest: (3, 15)            |       | −0.0303     |
| second smallest: (2, 10)     |       | −0.0302     |

TABLE 2. The ten largest  $|\phi|$  at  $x = 10^9$ , with the values after conditioning on  $\omega(p - 1)$  and on the fine signature  $\text{sig}(p)$  (pair-weighted mean of within-cell  $\phi$ ; cells with fewer than 200 primes excluded). “Dep.” marks multiplicatively dependent pairs.

| $a$ | $b$ | $\phi$  | $\phi \mid \omega$ | $\phi \mid \text{sig}$ | Dep. |
|-----|-----|---------|--------------------|------------------------|------|
| 5   | 13  | +0.0667 | +0.0200            | +0.0003                |      |
| 5   | 17  | +0.0649 | +0.0218            | +0.0025                |      |
| 5   | 29  | +0.0631 | +0.0246            | +0.0007                |      |
| 5   | 10  | +0.0624 | +0.0285            | −0.0125                | ✓    |
| 5   | 6   | +0.0623 | +0.0292            | +0.0005                |      |
| 5   | 11  | +0.0623 | +0.0300            | +0.0004                |      |
| 5   | 15  | +0.0622 | +0.0297            | −0.0255                | ✓    |
| 3   | 5   | +0.0622 | +0.0284            | +0.0015                | ✓    |
| 5   | 7   | +0.0622 | +0.0297            | +0.0002                |      |
| 2   | 5   | +0.0621 | +0.0336            | +0.0004                | ✓    |

One convention should be stated, since it affects the conditional averages in Sections 5 and 6. Within-cell  $\phi$  is undefined when a cell is constant in one of the two indicators (for example cells with  $5 \mid p - 1$ , in which no prime is Artin base 5). Such cells carry zero conditional covariance and are omitted from the pooled average, which therefore draws on 73% of primes on average for the signature stratification, and 23% for the finer signature-plus-character stratification, where three of the four character quadrants are identically non-Artin. Pooling instead on the covariance scale over all cells, which requires no such exclusion, gives 95.3% removal, in agreement with Table 3.

Three observations, in increasing order of depth.

**Observation 9** (Positivity).  $\phi(a, b) > 0$  for 64 of the 66 pairs. Artin statuses of different bases at the same prime *attract*.

The sign is the opposite of the consecutive-prime correlation measured in the first two papers of the series [1, 2] (where  $\delta < 0$  uniformly), and the mechanism is transparent: there the two primes had *different* (and LOS-anticorrelated) values of  $p - 1$ ; here every base is tested against the *same*  $p - 1$ . If  $p - 1$  has few small prime factors and a small 2-part, the primitive-root

TABLE 3. Mean cross-base correlation after conditioning, all 66 pairs at  $x = 10^9$ . “Removed” is  $1 - |\phi_{\text{cond}}|/|\bar{\phi}|$ , the means taken signed over the 66 pairs; on the mean-absolute scale the corresponding figures are 51% and 87%, and on the covariance scale 57% and 95%.

| conditioning                                           | mean $\phi$ | removed |
|--------------------------------------------------------|-------------|---------|
| none                                                   | 0.0352      | —       |
| $\omega(p-1)$                                          | 0.0158      | 55%     |
| $\text{sig}(p)$ : $(v_2, \{q \leq 13\}, \omega_{>13})$ | 0.0019      | 95%     |

condition is easier for every base at once. Formally, the Artin indicators share the common cause  $\text{sig}(p)$ .

**Observation 10** (The negative pairs are the entangled ones). The only pairs with  $\phi < 0$  are  $(2, 10)$  and  $(3, 15)$ : in each, the product of the two bases is, up to a square, the third base of a multiplicative triple contained in the set  $(2 \cdot 10 = 4 \cdot 5, 3 \cdot 15 = 9 \cdot 5)$ . For  $(2, 10)$ :  $P(A_{10} \mid A_2) = 0.3551$  versus  $P(A_{10} \mid \neg A_2) = 0.3853$ .

The sign reversal is Theorem 3 operating statistically. For every pair, dependent or not, the primes split into two halves according to the sign of  $\chi_d(p)$ ,  $d = \text{sqf}(ab)$ : on the half with  $\chi_d(p) = -1$  the values  $\chi_a(p)$  and  $\chi_b(p)$  are forced *opposite*, so joint Artin status is impossible there, while on the half with  $\chi_d(p) = +1$  they are forced *equal* and joint Artin status is unobstructed. This is visible directly in the data: over all 66 pairs and all 50,847,524 primes, the number of primes Artin for both bases with  $\chi_a(p) \neq \chi_b(p)$  is exactly zero, while the conditional joint density on the complementary half is 0.299.

What then distinguishes the two negatively correlated pairs is not the existence of the exclusion — every one of the 66 pairs has it, on exactly half the primes — but the identity of the barring character  $\chi_d$ , and in particular how its conductor interacts with the common cause  $p-1$ . Among the fifteen multiplicatively dependent pairs,  $d = \text{sqf}(ab)$  takes the values  $\{2, 3, 5, 6, 7, 10, 15, 21\}$ , and the two pairs with  $d = 5$  —  $(2, 10)$  and  $(3, 15)$ , of conductor 5, the smallest conductor occurring and the only odd prime one — are exactly the two with  $\phi < 0$ . That is the sharp empirical discriminant: not dependence, which thirteen positively correlated pairs also enjoy, but  $\text{cond}(\text{sqf}(ab)) = 5$ . A plausible reading is that a small conductor makes the barring condition  $\chi_d(p) = -1$  resolve on a coarse residue class that is comparatively well correlated with the divisibility structure of  $p-1$  driving the common cause, so the exclusion survives the averaging instead of being diluted by it; we do not have a proof of this and record it as an observation. Section 6 tests the reading by conditioning the common cause away, after which the exclusion becomes visible on six dependent pairs rather than two.

## 5. DECOMPOSITION: THE CORRELATION IS THE STRUCTURE OF $p-1$

Conditioning on  $\omega(p-1)$  alone — the crude size of the factorisation — removes barely half of the correlation. This is worth emphasising because  $\omega$  is the statistic through which the consecutive-prime papers routed the mechanism, and it is genuinely insufficient here: two primes with the same  $\omega(p-1)$  can present very different Artin conditions (it matters *which* primes divide  $p-1$ , and with what 2-adic valuation, not how many). The fine signature — 2-adic valuation capped at 4, the exact set of primes  $\leq 13$  dividing  $p-1$ , and the count of larger prime factors capped at 3 — is the information the Artin criterion actually consults for small bases, and conditioning on it removes 95% of the mean correlation (Table 3). That the fine

TABLE 4. Residual correlation after conditioning on  $\text{sig}(p)$  and after additionally conditioning on the pair of quadratic-residue statuses  $(\chi_a(p), \chi_b(p))$ , split by multiplicative dependence. All values at  $x = 10^9$  over 50,847,524 primes.

| group                                   | mean $\phi$ | mean $\phi \mid \text{sig}$ | mean $\phi \mid \text{sig, QR}$ |
|-----------------------------------------|-------------|-----------------------------|---------------------------------|
| multiplicatively dependent (15 pairs)   | +0.0314     | +0.0049                     | +0.0049                         |
| mean <i>absolute</i> residual           |             | 0.0179                      |                                 |
| multiplicatively independent (51 pairs) | +0.0363     | +0.0010                     | +0.0052                         |
| mean <i>absolute</i> residual           |             | 0.0010                      |                                 |
| all 66 pairs                            | +0.0352     | +0.0019                     | +0.0052                         |

structure of  $p - 1$ , rather than its coarse size, is what governs Artin behaviour is in the same spirit as the refined order-distribution conjectures of [3], which pass from the Artin condition  $\text{ord}_p(a) = p - 1$  to the distribution of  $\omega((p - 1)/\text{ord}_p(a))$  — finer information about the index than its mere vanishing, just as our signature is finer information about  $p - 1$  than  $\omega(p - 1)$ .

We regard this as the main statistical finding: *cross-base Artin correlation at a single prime is, to first order, not an interaction between the bases at all*. It is a common-cause effect of the shared integer  $p - 1$ . Any model that treats  $A_a$  and  $A_b$  as conditionally independent given the small-prime structure of  $p - 1$  reproduces 95% of the observed association.

## 6. THE RESIDUAL IS THE EXCLUSION LAW

The 5% that survives is not noise, and it is not uniformly distributed: it sits precisely on the multiplicatively dependent pairs. This section identifies it as the statistical residue of Theorem 3 — the deterministic exclusion that the unconditional correlation of Section 4 had largely averaged away.

Two facts pin down the mechanism, both measured over the full  $10^9$  range.

First, after fine conditioning the dependent pairs retain substantial residuals, and every pair with a substantial *negative* residual is multiplicatively dependent. The six pairs below are the only ones with  $\phi \mid \text{sig} < -0.004$ , and all six are dependent (Fisher exact  $p = 5.5 \times 10^{-5}$ ):

$$\begin{aligned} (2, 10): -0.0343, & \quad (5, 15): -0.0255, & \quad (6, 10): -0.0144, \\ (5, 10): -0.0125, & \quad (7, 21): -0.0060, & \quad (10, 15): -0.0049. \end{aligned}$$

At  $x = 10^9$  these are also the only six negative residuals of any size, but that sharper statement is not stable with  $x$ : at  $x = 3 \times 10^7$  a further twelve pairs have residuals marginally below zero, all within three standard errors of it.

The residual is not sign-definite, and we record the exceptions explicitly. Nine of the fifteen dependent pairs have *positive* residuals: eight of them small, and one,  $(2, 6)$ , a large outlier at +0.1598 — the largest residual of any pair in the collection. Its mechanism is transparent and is the same entanglement phenomenon seen from the other side:  $\text{sqf}(6) = 2 \cdot 3$ , so whenever  $3 \nmid p - 1$  the Artin conditions for 2 and for 6 very nearly coincide, and the signature records whether  $3 \mid p - 1$  without recording the resulting near-determinism inside those cells. (The effect is sharpest in the cells where  $p - 1$  has no small odd prime factor at all: for  $v_2(p - 1) = 1$ , no prime  $q \leq 13$  dividing  $p - 1$  and exactly one larger factor, the  $2 \times 2$  table for  $(2, 6)$  is exactly diagonal over 1,775,686 primes, giving  $\phi = +1$ . Across the 44 cells with  $v_2(p - 1) = 1$  and  $3 \nmid p - 1$  the within-cell  $\phi$  ranges from +0.39 to +0.98, so the near-determinism is a feature of that family of cells rather than of a single one.) The dependent-group mean absolute residual of 0.0179 is substantially driven by this one pair: excluding it, the figure is 0.0077, still eightfold

the independent-pair value of 0.0010. Conditioning additionally on  $(\chi_a(p), \chi_b(p))$  reduces (2, 6) to +0.0055, in line with every other pair — so the outlier is explained by the entanglement account rather than in tension with it. Note that (2, 10) and (5, 15) are more negative *after* conditioning than before, and that (5, 10), (5, 15), (6, 10), (10, 15) and (7, 21) change sign under conditioning: the common-cause channel had been masking an underlying repulsion.

Second, additionally conditioning on the pair of quadratic-residue statuses  $(\chi_a(p), \chi_b(p))$  removes this residual completely. Every one of the six negative values above becomes positive and small (+0.0031, +0.0066, +0.0045, +0.0065, +0.0040, +0.0048 respectively), and the dependent and independent groups become indistinguishable at +0.0049 and +0.0052 (Table 4). That common small positive floor of about +0.005 is present for all 66 pairs and is plausibly attributable to higher-order (cubic and quartic) conditions shared through  $p - 1$  beyond what our signature records.

Conditioning on the quadratic characters is precisely what removes character-level entanglement, and that it flattens the dependent pairs identifies the residual as the statistical trace of the relation  $\chi_a \chi_b = \chi_{\text{sqf}(ab)}$  — the same object that makes Theorem 3 true.

Indeed the identification can be made exactly, and not merely statistically. Theorem 3 asserts that the joint Artin event is empty on the stratum  $\chi_a(p) \neq \chi_b(p)$ , and this is what the contingency tables show: summing the joint-Artin cell over all signature classes and all 66 pairs, the count of primes  $p \leq 10^9$  that are Artin for both  $a$  and  $b$  with  $\chi_a(p) \neq \chi_b(p)$  is

$$0 \quad (\text{out of } 50,847,524 \text{ primes, for every one of the } 66 \text{ pairs}),$$

while on the complementary stratum the joint density is 0.299. The residual measured in this section is therefore not evidence *for* a conjectural mechanism: the mechanism is a theorem, and what the residual measures is how much of that deterministic constraint survives the averaging over  $p - 1$  that the unconditional correlation performs.

One caveat on the strength of the flattening step, in the other direction. Within a  $(\chi_a, \chi_b)$  stratum the exclusion cannot operate at all: by criterion (1) an Artin prime for  $a$  has  $\chi_a(p) = -1$ , so all the Artin mass sits in the single quadrant  $(-1, -1)$ , and the remaining three quadrants contribute no variation in either indicator. Conditioning on  $(\chi_a, \chi_b)$  therefore *restricts attention to the stratum on which the constraint of Theorem 3 is vacuous*, and the vanishing of the residual there is in part a structural consequence of that restriction rather than an independent confirmation. What the step establishes is the localisation: the residual lives in the character channel and not elsewhere in the signature. We state it in that weaker form.

The picture that emerges is cleanly layered:

$$\underbrace{\text{common cause } p-1}_{\approx 95\%} + \underbrace{\text{quadratic entanglement } \chi_a \chi_b = \chi_{\text{sqf}(ab)}}_{\text{residual, localised on dependent pairs}} + \underbrace{\text{higher-order sharing}}_{\text{small positive floor}}.$$

## 7. COMPARISON WITH THE PREDICTED JOINT DENSITIES

The measurements of Sections 4–6 identify the mechanisms empirically. We now check them against theory quantitatively, by computing the joint densities predicted by the Matthews–Moree–Stevenhagen framework and comparing with the observed counts pair by pair.

Write  $d_a = \text{sqf}(a)$ . The heuristic, in the form originating with [11] and systematised by the character-sum method of [10] (with the general framework in [5]), is an inclusion–exclusion over the degrees of the Kummer extensions that detect the conditions “ $a$  is an  $\ell$ -th power residue”:

$$\mathbf{P}(p \in A_a \cap A_b) = \sum_{m,n \text{ squarefree}} \frac{\mu(m) \mu(n)}{[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}]}, \quad L = \text{lcm}(m, n), \quad (3)$$

TABLE 5. Predicted versus measured joint Artin densities, selected pairs at  $x = 10^9$ . “Dep.” marks multiplicatively dependent pairs. The final two rows are the two negatively correlated pairs.

| $a$ | $b$ | measured $j$ | model $j$ | rel. err. | measured $\phi$ | model $\phi$ | Dep. |
|-----|-----|--------------|-----------|-----------|-----------------|--------------|------|
| 5   | 13  | 0.163939     | 0.163943  | +0.00%    | +0.0667         | +0.0667      |      |
| 5   | 10  | 0.161962     | 0.161922  | −0.02%    | +0.0624         | +0.0623      | ✓    |
| 5   | 15  | 0.161934     | 0.161922  | −0.01%    | +0.0622         | +0.0623      | ✓    |
| 3   | 7   | 0.149965     | 0.149971  | +0.00%    | +0.0432         | +0.0433      | ✓    |
| 13  | 17  | 0.150249     | 0.150261  | +0.01%    | +0.0383         | +0.0383      |      |
| 2   | 3   | 0.147361     | 0.147349  | −0.01%    | +0.0321         | +0.0321      | ✓    |
| 2   | 6   | 0.147334     | 0.147349  | +0.01%    | +0.0320         | +0.0321      | ✓    |
| 7   | 21  | 0.144754     | 0.144728  | −0.02%    | +0.0238         | +0.0238      | ✓    |
| 15  | 21  | 0.144725     | 0.144728  | +0.00%    | +0.0237         | +0.0238      |      |
| 3   | 15  | 0.132769     | 0.132776  | +0.01%    | −0.0303         | −0.0302      | ✓    |
| 2   | 10  | 0.132776     | 0.132776  | +0.00%    | −0.0302         | −0.0302      | ✓    |

and the entanglement enters through the degree in the denominator. Generically

$$[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}] = \varphi(L) m n,$$

but the degree drops by the order of the subgroup of quadratic classes realised inside  $\mathbb{Q}(\zeta_L)$ , taken from the group

$$V(m, n) = \langle d_a \text{ (if } 2 \mid m), d_b \text{ (if } 2 \mid n) \rangle \subseteq \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$$

whose quadratic field lies inside  $\mathbb{Q}(\zeta_L)$ , i.e. whose conductor divides  $L$ . The crucial point for this paper is that when both  $2 \mid m$  and  $2 \mid n$  the group  $V$  contains not only  $d_a$  and  $d_b$  but their product  $d_a d_b \sim \text{sqf}(ab)$ : the barring character  $\chi_{\text{sqf}(ab)}$  of Theorem 3 appears in the model as an extra degree collapse, exactly for the multiplicatively dependent pairs. We evaluate (3) by summing the ramified part exactly over all squarefree  $m, n$  supported on  $S = \{2\} \cup \{q : q \mid d_a d_b\}$  and folding the rest into the Euler product  $\prod_{q \notin S} (1 - \frac{2q-1}{q^2(q-1)})$ , truncated at  $3 \times 10^6$ .

As a check on the implementation, the same code specialised to a single base reproduces the classical Hooley densities: 0.373956 for bases with  $d \not\equiv 1 \pmod{4}$  (Artin’s constant), 0.393638 for  $d = 5$ , 0.376368 for  $d = 13$ , against measured 0.373957, 0.393642, 0.376362.

**Observation 11** (The model reproduces the measurements). Over all 66 pairs at  $x = 10^9$ , the mean absolute relative error between the predicted and measured joint densities is 0.012%, with maximum 0.040% (attained at (21, 29)). In particular the model predicts the sign reversal: for the entangled pairs (2, 10) and (3, 15) it gives  $j = 0.132776$  and  $\phi = -0.0302$  for both, against measured  $j = 0.132776, 0.132769$  and  $\phi = -0.0302, -0.0303$  respectively.

This is the quantitative counterpart of Sections 4 and 6, and it settles the interpretation. The 66-pair pattern — positive correlation almost everywhere, negative precisely on the two entangled pairs, with magnitudes ordered as observed — is not merely *consistent with* the Matthews–Moree–Stevenhagen framework; it is reproduced by it to within a few parts in  $10^4$ , with no fitted parameters. Correspondingly, the deterministic law of Theorem 3 and the statistical residual isolated in Section 6 are two faces of one object: the character  $\chi_{\text{sqf}(ab)}$ , which bars half the primes pointwise and contributes the degree collapse in  $V(m, n)$  asymptotically.

**Remark 12.** The residual discrepancies are uniformly small (all 66 pairs below 0.040%) and we have not pursued them; ordinary Hooley-type errors of this size are expected at  $x = 10^9$ , since the convergence in Hooley’s theorem is only logarithmic, and our Euler products are truncated at  $3 \times 10^6$ . We record one methodological warning, since it cost us a draft. Computing  $\varepsilon(m, n)$  as 2 raised to the number of classes of  $V(m, n)$  with conductor dividing  $L$  — rather than as the order of the subgroup of  $V(m, n)$  realised in  $\mathbb{Q}(\zeta_L)$  — inflates  $\varepsilon$  to 8 on exactly those pairs with  $d_a \equiv d_b \equiv 1 \pmod{4}$ , where all three of  $d_a, d_b, \text{sqf}(ab)$  have odd conductor. Since  $V(m, n)$  has two generators,  $|V(m, n)| \leq 4$  and  $\varepsilon = 8$  is impossible. The error is invisible in the single-base specialisation, which is why the check above does not catch it; it inflates the mean relative error over the 66 pairs from 0.012% to 0.037% and concentrates the damage on the ten pairs drawn from  $\{5, 13, 17, 21, 29\}$ .

## 8. DISCUSSION

**Relation to the first two papers of the series** [1, 2]. The trilogy now covers both axes of the joint distribution of Artin status. Along the prime axis (consecutive  $p_n, p_{n+1}$ , one base), correlation is negative, driven by the Lemke Oliver–Soundararajan bias [9] propagating through the two distinct integers  $p_n - 1$  and  $p_{n+1} - 1$ , with deterministic exclusion classes in the gap. Along the base axis (one prime, two bases), correlation is positive, driven by the single shared integer  $p - 1$ , with deterministic exclusion on an explicit half of all primes for every pair of bases. In both cases the deterministic law is a parity statement about quadratic characters attached to a congruence condition — there on the gap  $p_{n+1} - p_n$ , here on  $p$  itself — and in both cases it is statistically dominated by a factorisation-of- $p - 1$  mechanism: diluted to invisibility in the global average there, absorbed almost entirely by conditioning here. The symmetry is closer than we expected when the series began: what looked like a statement about special configurations of bases (multiplicative triples) turned out, once the third base was seen to be inessential, to be a statement of the same shape as the gap law — a character condition on a residue class of density  $1/2$ .

**Future work.** (1) A rigorous version of Section 7: the comparison there is between measurement and a heuristic evaluated numerically, and the error terms — both the Hooley-type error at  $x = 10^9$  and the truncation of our Euler tail — are not controlled. Making the agreement into a theorem (conditionally on GRH, in the manner of [11]) would require uniform error bounds we do not attempt. (2) The conductor dependence made precise. Theorem 3 bars half the primes for every pair of our base set, yet only the two pairs with  $\text{cond}(\text{sqf}(ab)) = 5$  show a negative unconditional correlation; Section 4 offers a heuristic reason but no proof. Quantifying the surviving exclusion as a function of  $\text{cond}(\text{sqf}(ab))$  — equivalently, computing the covariance between the barring condition  $\chi_{\text{sqf}(ab)}(p) = -1$  and the signature of  $p - 1$  — looks tractable within the Kummer-degree framework and would turn the empirical discriminant into a formula. A natural test is to enlarge the base set so that several small odd prime conductors occur among the dependent pairs, which our twelve bases do not provide. (3) The mixed axis:  $\phi$  between  $A_a(p_n)$  and  $A_b(p_{n+1})$  for  $a \neq b$  — the data pipeline extends immediately, and the prediction from the present findings is a small negative correlation governed by the two conductors jointly. (4) Higher-order entanglement: triples of bases conditioned on quadratic data should expose the cubic layer ( $\ell = 3$  Kummer conditions), with  $\mathbb{Q}(\zeta_3)$  playing the role  $\mathbb{Q}(i)$  and  $\mathbb{Q}(\sqrt{2})$  played here. The analogue of Theorem 3 at  $\ell = 3$  would bar  $2/3$  of primes on a cubic-residue condition, and we do not know whether it is visible above the common cause.

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All computations were executed and verified on the author’s own hardware from source code the author has inspected. The proofs of Theorems 3 and 4 have been checked by hand and machine-checked in Lean 4 against Mathlib with no appeal to any axiom beyond `propext`, `Classical.choice` and `Quot.sound`, including the step (1) and, for the instance  $(a, b) = (5, 10)$ , the discharge of the barring condition  $\left(\frac{2}{p}\right) = -1$  from  $p \bmod 8$  — so that instance is verified with no character hypothesis assumed at all. The general translation of  $\left(\frac{d}{p}\right) = -1$  into residue classes mod  $4d$  is not formalised generically; it is carried out by hand per instance. See `LEAN_NOTE.md` in the accompanying package. Every reference has been checked against the published record, with two stated exceptions: the companion preprints [1, 2] are unrefereed and have no published record, and [11] was verified from the publisher’s bibliographic record and from independent restatements of its main theorem rather than from the original text, which the author was unable to obtain. The author is solely responsible for the content.

## DECLARATION OF INTEREST

The author reports there are no competing interests to declare.

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## DATA AVAILABILITY

All code and results are publicly available at [repositoryURLwithheldforreview](#), including the single-pass multi-base programs (`crossbase.c`, `crossbase_fine.c`, `crossbase_om.c`, `crossbase_qr.c`), the analysis scripts, and the raw contingency-table output at  $x = 10^9$  from which every value quoted here is derived, namely

`crossbase_qr_1e9.json`, `crossbase_fine_1e9.json`, `crossbase_om_1e9.json`

(3.2 MB in total). Running `analyze_qr.py` on the first of these reproduces Table 4 exactly. The full computation reproduces in under two hours on a single core; the  $x = 10^9$  signature-and-character run alone takes about 16 minutes on 1 core of a modern desktop. The Lean sources for the machine-checked statements are included in the same repository, with build instructions in `BUILD.md`.

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